1. Clinical Question & Core Hypothesis

Clinical Question

Why do some pretreatment TNBCs with high stromal TILs still fail to achieve pCR after NAC? The working mechanistic question is whether immune presence (sTILs) is only beneficial when those lymphocytes are functionally activated (IFN-γ signaling) and not counterbalanced by adaptive immunosuppression?

Working Hypothesis

In pretreatment TNBC, high stromal TIL density predicts higher likelihood of achieving pathological complete response (pCR) to neoadjuvant chemotherapy only when IFN-γ signaling is functionally active and IDO1–mediated immunosuppression is minimal, such that sTIL-high tumors with high IFNG and low IDO1 exhibit a robust cytotoxic/inflammatory program conducive to pCR. An integrated biomarker combining sTIL density with IFNG and IDO1 expression will more accurately predict pCR than sTIL density alone by capturing the balance between effector activation (and associated Tier 2 signals such as CXCL9/10, GZMB, PRF1, CD8A, HLA-DRA) and immunosuppressive counter-regulation.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	IFNG, IDO1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

IFNG: IFNG reflects a coordinated effector program; their patterns can validate the mechanism linking IFNG to effector function and IDO1 to adaptive suppression.

IDO1: IDO1) because these directly explain functional competence vs.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CXCL9	Supporting modifier from knowledge graph
Secondary	CXCL10	Supporting modifier from knowledge graph

Secondary	GZMB	Supporting modifier from knowledge graph
Secondary	PRF1	Supporting modifier from knowledge graph
Secondary	CD8A	Supporting modifier from knowledge graph
Secondary	HLA-DRA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: quantitative sTIL assessment via standardized pathology or image analysis,

Prediction 2: targeted transcript/protein measurements for IFNG, IDO1, CXCL9/10, GZMB, PRF1, CD8A, HLA-DRA, and

Prediction 3: multivariate modeling (logistic regression or machine learning) to test whether the integrated biomarker outperforms sTIL alone in predicting pCR, with validation in an independent cohort. Confounders such as tumor purity, sampling bias, and treatment regimens should be controlled.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CXCL9: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CXCL10: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.